Hae Seong Lee

Postdoctoral Researcher in Statistical Physics

Statistical Physics of Eco-Evolutionary Dynamics (SPEED) Lab, Department of Physics, Inha University

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Research Interests

- Emergent phenomena in complex systems
- Dynamics in complex networks
- Ecological Systems
- Synchronization

Research Experiences

Inha University — *Incheon, South Korea*

Postdoctoral Researcher — *Mar. 2025 - Present*

- Host: Prof. Hye Jin Park
- NRF postdoctoral fellow (Sep. 2026 – Aug. 2027)
- Random Lotka–Volterra dynamics on complex networks: how degree-correlated interaction strengths reshape biodiversity and network assortativity (PRE, under review)
- Eco-evolutionary model of migratory timing under seasonal resources and competition

Education

Sungkyunkwan University — *Suwon, South Korea*

Ph.D. in Physics — *Mar. 2019 - Feb. 2025*

- Thesis: Clap, Sing, and Sleep: A Study of Emergent States in the Kuramoto Model
- Supervisor: Prof. Beom Jun Kim

Sungkyunkwan University — *Suwon, South Korea*

B.S. in Physics — *Mar. 2013 - Aug. 2017*

Publications

Published

[5] **Hae Seong Lee**, Deok-Sun Lee, Sang Hoon Lee, Samir Suweis, and Hye Jin Park, "Ecological networks of viable species with degree-dependent interaction", Phys. Rev. E 114, 054307 (2026).

[4] Younghwa Cha, Yeji Lee, Eunhee Ji, SoHyun Han, Sunhyun Min, Hyoungkyu Kim, Minseo Cho, **Hae Seong Lee**, Youngjai Park, and Joon-Young Moon, "Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI", Sci. Data (2026)

[3] Jae Hyung Woo, **Hae Seong Lee**, Tae-Wook Ko and Joon-Young Moon, "Hysteresis in a Generalized Kuramoto Model with a First-order Approximate Coupling Function and Inhomogeneous Coupling Strengths", Chaos, Solitons and Fractals 190, 115770 (2025).

[2] **Hae Seong Lee**, Beom Jun Kim, and Hye Jin Park, "Stability of Twisted States in Power-law-coupled Kuramoto Oscillators on a Circle with and without Time Delay", Phys. Rev. E 109, 064203 (2024).

[1] **Hae Seong Lee**, Jong Il Park, and Beom Jun Kim, "Modified Kuramoto Model with Inverse-square Law Coupling and Spatial Time Delay", Physica A 582, 126263 (2021).

Preprints

In preparation

- **Hae Seong Lee** and Beom Jun Kim, "The Effect of Prior Knowledge on Vocabulary Learning"
- **Hae Seong Lee**, Hyeong-Chai Jeong, Kalle Parvinen, Ulf Dieckmann, and Hye Jin Park, "Migration Patterns of Migratory Birds Driven by Seasonal Resource Variation and Competition"

Grants and Fellowships

Postdoctoral Research Fellowship
National Research Foundation of Korea (NRF)
Sep. 2026 - Aug. 2027

- Principal Investigator. Project: Power Grid Synchronization Stability under Renewable Energy Expansion: An Approach Based on the Diversity-Stability Paradox

Honors and Awards

SKKU Scholarship Innovation
Sungkyunkwan University
2022

SKKU Innovative Research Fellowship
Sungkyunkwan University
2022

Excellence Project Award
19th KIAS-APCTP Winter School on Statistical Physics
2022

Outstanding Oral Presentation Award
2021 Korean Physical Society Spring Meeting
2021

Graduate Merit Scholarship
Sungkyunkwan University
2019-2021

Skills

Languages	Korean(native), English(fluent)
Programming Languages	C, C++, Python, SQL, Matlab, Javascript
Computer Skills	Monte-carlo simulation, CUDA programming, Agent-based simulation, MPI programming, Networkx, Latex

Experience

Teaching Assistant

Thermal and Statistical Mechanics II
Sungkyunkwan University
Suwon, South Korea
Spring 2020, Spring 2021

Thermal and Statistical Mechanics I
Sungkyunkwan University
Suwon, South Korea
Fall 2020, Fall 2021

Analytical Mechanics II Exercise
Sungkyunkwan University
Suwon, South Korea
Fall 2019

Analytical Mechanics II
Sungkyunkwan University
Suwon, South Korea
Fall 2019

Analytical Mechanics I
Sungkyunkwan University
Suwon, South Korea
Spring 2019

Military service

Technical Research Personnel
Sungkyunkwan University
Suwon, South Korea
Mar. 2022 - Feb. 2025

Services

Organizer of the joint journal club of statistical physicists
Sungkyunkwan University, Hanyang University, Catholic University of Korea, and Inha University
South Korea
Aug. 2024 - Feb. 2025

Attended Schools

International Schools

Unifying Mathematical Models of Biodiversity
Topic: A theoretical overview of ecological dynamics and complexity
Tarascon-sur-Ariege, France
Jul. 2026

XII GEFENOL Summer School on Statistical Physics of Complex Systems
Topic: Selected topics in statistical physics
Madrid, Spain
Jul. 2024

Complexity72h
Topic: Unraveling cancer dynamics: From multiscale stochastic models to tissue morphology
Madrid, Spain
Jun. 2024

Complex Networks: Theory, Methods, and Applications
Topic: Selected topics in complex networks
Como, Italy
May 2024

CSH Winter School
Topic: Integrative and Disintegrative Processes in Complex Human Societies
Obergurgl, Austria
Jan. 2023

VII Mediterranean school on complex networks
Topic: Selected topics in complex networks
Catania, Italy
Jun. 2022

Domestic Schools

22nd KIAS-APCTP Winter School on Statistical Physics — *Pohang, South Korea*
Topic: Statistical physics of biological and soft matter — *Jan. 2025*

21st KIAS-APCTP Winter School on Statistical Physics — *Pohang, South Korea*
Topic: Selected topics in statistical physics on complex systems — *Jan. 2024*

Complex systems summer school — *Seoul, South Korea*
Topic: Lectures on "Network Science" written by Barabasi — *Jul. 2023*

20th KIAS-APCTP Winter School on Statistical Physics — *Pohang, South Korea*
Topic: Statistical Physics of Quantum Systems — *Jan. 2023*

19th KIAS-APCTP Winter School on Statistical Physics — *Online*
Topic: Phase transitions and critical phenomena — *Jan. 2022*

18th KIAS-APCTP Winter School on Statistical Physics — *Online*
Topic: Kinetic models in statistical physics — *Jan. 2021*

17th KIAS-APCTP Winter School on Statistical Physics — *Pohang, South Korea*
Topic: Machine learning for statistical physics in practice — *Jan. 2020*

16th KIAS-APCTP Winter School on Statistical Physics — *Yeosu, South Korea*
Topic: Critical phenomena and renormalization group — *Jan. 2019*

Presentations

International Oral Presentation

[9] "Migration Patterns of Migratory Birds Driven by Seasonal Resource Variation and Competition", RHINO 2026 (Yokohama, Japan, Aug. 30, 2026)

[8] "Species Survival Probability Under Degree-interaction Strength Correlation", Arctic Meeting for Adaptive Mechanisms in Biological Systems (Abisko, Sweden, Jan. 20, 2026)

[7] "Bird Migration Patterns Arising from Seasonal Resource Availability and Competition", Conference on Complex Systems 2025 (Siena, Italy, Sep. 1, 2025)

[6] "Hysteresis in Coupled Identical Oscillators with Generalized Coupling Function and Coupling Strength Inhomogeneity", Complex Networks 2023 (Menton, France, Nov. 28, 2023)

[5] "Stability of Twisted States in Kuramoto Oscillators on a Circle with Power-law Interaction Strength", Roles of heterogeneity in non-equilibrium collective dynamics (RHINO 2022) (Tokyo, Japan, Sep. 18, 2022)

[4] "Twisted States in Kuramoto Oscillators on a Circle with Distance-decaying Coupling Strength", 15th Asia Pacific Physics Conference (Online, Aug. 22, 2022)

[3] "Synchronization of Oscillators with Power-law Coupling and Spatial Time Delay", 2022 Korea-Canada Symposium and International Workshop on Multiplex Brain Networks (Calgary & Banff, Canada, Apr. 21, 2022)

[2] "The Central Word Set in a Language", Networks 2021 (Online, Jul. 5, 2021)

[1] “Modified Kuramoto Model with Power Law Coupling and Spatial Time Delay”, Conference on Complex Systems 2019 (Singapore, Sep. 30, 2019)

Domestic Oral Presentation

[9] “Degree-correlated Interactions Drive Assortativity in Ecological Networks through Hub-Favored and Hub-Suppressed Phases”, 2026 Korean Physical Society Spring Meeting (Daejeon, South Korea, Apr. 23, 2026)

[8] “Evolution of Risk-taking Strategy in von Neumann Poker Game”, Workshop on the game theory and its applications (Gwangmyeong, South Korea, Nov. 24, 2023)

[7] “Stability of Twisted States in Kuramoto Oscillators on a Circle with Distance-Decaying and Time-Delayed Coupling”, 2023 Korean Physical Society Spring Meeting (Daejeon, Apr. 20, 2023)

[6] “The Compartment Learning Model and the Core Vocabulary in Korean Language”, 2021 Korean Physical Society Spring Meeting (Online, Apr. 22, 2021)

[5] “A Set of Central Words in Korean Language”, 2020 Korea Academy of Complexity Studies Fall Conference (Online, Nov. 28, 2020)

[4] “A Set of Central Words in Korean Language”, Joint mini-workshop on collective dynamics (Busan, South Korea, Nov. 12, 2020)

[3] “A Set of Central Words in Korean Language”, 2020 Korean Physical Society Fall Meeting (Online, Nov. 5, 2020)

[2] “Modified Kuramoto Model with Power Law Coupling and Site-dependent Time Delay”, 2019 Korean Physical Society Fall Meeting (Gwangju, South Korea, Oct. 24, 2019)

[1] “Modified Kuramoto Model with Power Law Coupling and Spatial Time Delay”, The 20th Workshop for Statistical Physics (Byeonsan, South Korea, Aug. 21, 2019)

International Poster Presentation

[6] “How Degree-dependent Interactions Reshape Hub Survival in Ecological Networks”, Netsci 2026 (Boston, USA, Jun. 3, 2026)

[5] “Generalized Random Lotka-Volterra Model with Degree-correlated Interaction Strengths”, Statphys29 (Florence, Italy, Jul. 14, 2025)

[4] “How Do Degree-correlated Interaction Strengths Shape Biodiversity?”, ACMB-JSMB 2025 (Kyoto, Japan, Jul. 8, 2025)

[3] “Why We Sing in a Round at a Stadium: Stability Analysis of Twisted States in Kuramoto Oscillators on a Circle”, Netsci 2024 (Quebec, Canada, Jun. 17, 2024)

[2] “Finding Core Vocabulary in a Language”, Statphys28 (Tokyo, Japan, Aug. 8, 2023)

[1] “Effect of Prior Knowledge on Vocabulary Learning”, Conference on Complex Systems 2022 (Palma, Spain, Oct. 18, 2022)

Domestic Poster Presentation

[5] “Impact of Degree-correlated Interactions on Biodiversity”, 2025 Korean Physical Society Spring Meeting (Daejeon, South Korea, Apr. 23, 2025)

[4] “Biodiversity in Random Lotka-Volterra Model with Degree-correlated Interaction Strength”, Stochastic Processes as Tools for Understanding Biological Systems (Busan, South Korea, Feb. 17, 2025)

[3] “Locally Twisted States of Kuramoto Oscillators in a Brain Network with Site-dependent Time Delay”, 2024 Korean Physical Society Fall Meeting (Yeosu, South Korea, Oct. 25, 2024)

[2] “Finding Important Words: Network Analysis of a Korean Dictionary”, 2024 Korean Physical Society Spring Meeting (Daejeon, South Korea, Apr. 25, 2024).

[1] “Emergence of Twisted States in Kuramoto Oscillators on a Circle with Power-law Coupling

Strength”, 2022 SKKU Physics Workshop (Suwon, South Korea, Nov. 30, 2022)